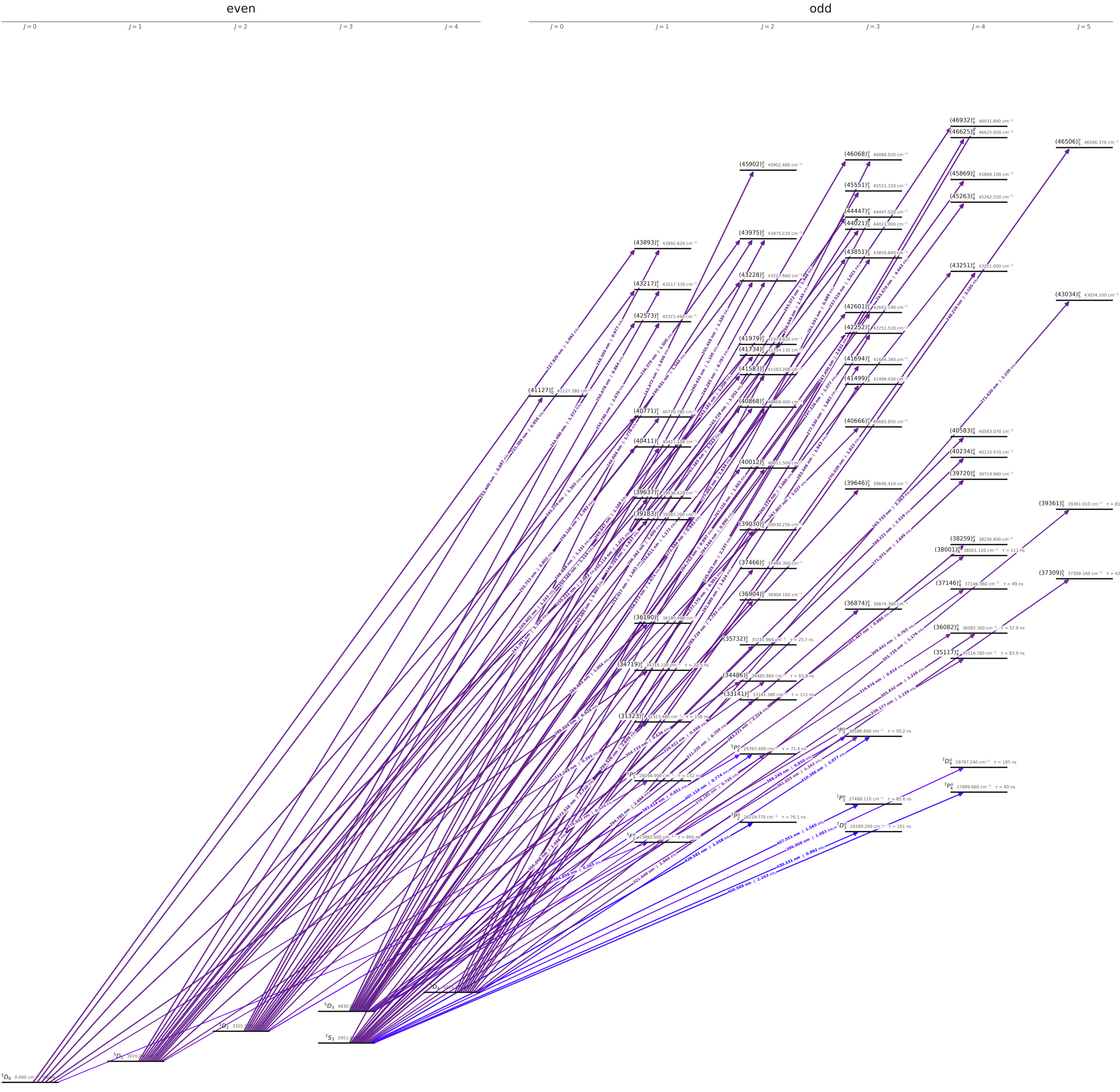


W<sup>+</sup> ionisation limit 63427.70 cm<sup>-1</sup> = 7.86403 eV (157.660 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $\langle 0 || er || J' \rangle$  in  $a_0$   
 Matrix elements are measured or NIST values; \* = high-accuracy theory, = model calculation.  
 Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation,  $n = 70$ .  
 Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



| Transition        | $\lambda$ vac (nm) | $\lambda$ air (nm) | $\nu$ (THz) | $ \langle J  \mathbf{J}  J\rangle $ (ea) | A (s <sup>-1</sup> ) | $\Gamma/2\pi$ up. (MHz) | branch (%) | Use                                                          |
|-------------------|--------------------|--------------------|-------------|------------------------------------------|----------------------|-------------------------|------------|--------------------------------------------------------------|
| $^5D_0 - ^5F_2^o$ | 384.851            | 384.749            | 778.9687    | 0.265                                    | 8.333 $\pm$ 0.5      | 0.1851                  | 72.4       | precision laser spectroscopy: isotope shifts and 183W hyp... |
| $^7S_3 - ^7P_2^o$ | 400.9882           | 400.8749           | 747.6341    | 2.163                                    | 1.633 $\pm$ 0.7      | 2.653                   | 98.3       | 400.9 nm line: tungsten-influx                               |
| $^7S_3 - ^7P_2^o$ | 407.5508           | 407.4357           | 735.5954    | 1.565                                    | 1.048 $\pm$ 0.7      | 1.859                   | 89.7       | precision laser spectroscopy: isotope shifts and 183W hyp... |

$\lambda$ ,  $\nu$  from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.  $\Gamma/2\pi$ : natural linewidth of the upper level,  $1/(2\pi\tau)$ .

| Level       | A (MHz)     | B (MHz) | Shift of each F level from the centre of gravity (MHz) |
|-------------|-------------|---------|--------------------------------------------------------|
| ${}^2S_3$   | 505.592(12) | -       | F=5/2: -1011.18 F=7/2: +758.39                         |
| $5F_4^o$    | 47.1(11)    | -       | F=1/2: -47.10 F=3/2: +23.55                            |
| ${}^2P_3^o$ | 496.2(6)    | -       | F=5/2: -992.40 F=7/2: +744.30                          |
| ${}^2P_4^o$ | 440.5(6)    | -       | F=7/2: -1101.25 F=9/2: +881.00                         |

Measured A, B constants; sources in the data file.

| Transition      | $\lambda$ vac (nm) | Isotopes | Shift (MHz) |
|-----------------|--------------------|----------|-------------|
| $^5D_0 - ^5F_1$ | 384.858            | 186-184  | -12.7(16)   |
| $^5D_0 - ^5F_1$ | 384.858            | 184-182  | -10.0(16)   |
| $^5D_0 - ^5F_1$ | 384.858            | 183-182  | -4.5(16)    |
| $^7S_3 - ^7P_2$ | 400.988            | 186-184  | 423.6(21)   |
| $^7S_3 - ^7P_2$ | 400.988            | 184-182  | 498.4(21)   |
| $^7S_3 - ^7P_2$ | 400.988            | 183-182  | 263.2(21)   |
| $^7S_3 - ^7P_2$ | 400.988            | 182-180  | 334.0(21)   |
| $^7S_3 - ^7P_2$ | 407.551            | 186-184  | 463.6(29)   |
| $^7S_3 - ^7P_2$ | 407.551            | 184-182  | 541.4(29)   |
| $^7S_3 - ^7P_2$ | 407.551            | 183-182  | 286.9(29)   |
| $^7S_3 - ^7P_2$ | 407.551            | 182-180  | 355.6(29)   |

Shift =  $v$ first (isotope) -  $v$ second (isotope).

$$\text{Shift} = \nu(\text{first isotope}) - \nu(\text{second isotope}).$$